

Google Cloud Professional Cloud Architect (PCA)

Comprehensive 40-Question Practice Exam Bank • 8 Core Exam Domains

Topic 1: PCA Exam Objectives & Case Studies

Question 1.1

Case Study Focus: EHR Healthcare

EHR Healthcare is migrating its customer-facing SaaS electronic health record applications from expiring colocation facilities to Google Cloud. The executive team mandates a minimum 99.9% availability SLA across customer-facing systems, continuous software release deployment, reduced global latency for medical offices, and minimized administrative overhead. Which architecture best meets these business and technical requirements?

- A. Host containerized workloads on Compute Engine virtual machines in a single region, managing cluster orchestration manually with custom scripts.
- B. Deploy regional Google Kubernetes Engine (GKE) clusters across multiple Google Cloud regions behind a global Application Load Balancer using Premium Network Tier, paired with managed database services.
- C. Deploy a single zonal GKE cluster in us-central1, utilizing Standard Network Tier for egress and manual nightly database backups.
- D. Migrate containerized applications to Spot VMs on Compute Engine with local SSDs, relying on email alerts for system health.

Correct Answer: B

Explanation: Regional GKE clusters deployed across multiple regions behind a global Application Load Balancer ensure high availability exceeding the 99.9% SLA target. Utilizing Google's Premium Network Tier minimizes global latency, while managed database services reduce infrastructure administration costs.

Question 1.2

Case Study Focus: Altostrat Media

Altostrat Media maintains legacy on-premises workflows for media ingestion and archival while managing an expanding library of audio, video, and document assets in Cloud Storage. To support modernizing content ingestion, they need secure, high-performance hybrid connectivity and a solution to optimize cloud storage costs for growing media archives. What should the Cloud Architect recommend?

- A. Connect the on-premises datacenter via Cloud Interconnect for high-speed ingestion, and apply Cloud Storage Object Lifecycle Management rules to transition older assets to Coldline or Archive storage.
- B. Ingest media over public internet HTTP endpoints and store all files permanently in the Standard Cloud Storage class.
- C. Connect on-premises datacenters using VPC Network Peering and store media assets as JSON blobs inside BigQuery tables.
- D. Use SFTP batch file transfers over an unencrypted connection and manually delete legacy assets after 30 days.

Correct Answer: A

Explanation: Dedicated or Partner Cloud Interconnect provides the secure, high-performance hybrid connection required for continuous high-volume data ingestion. Cloud Storage Object Lifecycle Management automatically shifts infrequently accessed media files to lower-cost storage classes (Coldline/Archive), satisfying the cost optimization mandate.

Question 1.3

Case Study Focus: Cymbal Retail

Cymbal Retail wants to enhance product discoverability and conversational commerce by allowing customers to search for products using natural language on their website and mobile app. Additionally, retail associates require a mechanism to review, edit, or approve AI-generated catalog attributes before they are published live. Which combination of Google Cloud capabilities addresses these needs?

- A. Integrate Google Cloud's Discovery AI for natural language product retrieval, and deploy a custom Human-in-the-Loop (HITL) review UI for retail associates.
- B. Replace all relational databases with Firestore and run automated SQL scripts inside BigQuery ML without human intervention.
- C. Route customer natural language queries to Interactive Voice Response (IVR) call center agents for manual catalog search.
- D. Store catalog attributes in flat CSV files on Cloud Storage FUSE and send email notifications to associates.

Correct Answer: A

Explanation: Cymbal Retail's solution concept specifically requires integrating Google Cloud Discovery AI to process natural language user requests for product discovery. The technical requirements mandate a Human-in-the-Loop (HITL) review UI allowing staff to review and edit AI-generated attributes before catalog updates.

Question 1.4**Case Study Focus: KnightMotives Automotive**

KnightMotives is expanding its autonomous vehicle development program into international regions, including the European Union. They must strictly enforce EU data protection regulations (GDPR), control data residency, prevent telemetry data exfiltration, and protect against security breaches. How should the architect secure the cloud data environment?

- A. Store telemetry data in public Cloud Storage buckets to simplify access for international testing teams.
- B. Establish VPC Service Controls perimeters around sensitive data APIs, enforce Customer-Managed Encryption Keys (CMEK) via Cloud KMS, and enforce location constraints via Organization Policies.
- C. Stream telemetry directly over unencrypted HTTP endpoints into a single Compute Engine instance using static service account keys.
- D. Rely on legacy SFTP batch scripts to transfer vehicle logs to an on-premises mainframe without perimeter security.

Correct Answer: B

Explanation: VPC Service Controls prevent data exfiltration by restricting access to Google Cloud service APIs. Cloud KMS CMEK ensures cryptographic key governance, and Organization Policy location constraints enforce strict data residency compliance with EU regulations.

Question 1.5**Case Study Focus: Altostrat Media**

Altostrat Media needs to process uploaded media files by automatically generating concise summaries, extracting metadata using NLP, and filtering harmful or inappropriate content. Their executive statement highlights that all AI decision-making must be auditable and explainable. Which architecture fulfills these requirements?

- A. Execute serverless event-driven processing using Cloud Run functions, call Vertex AI foundation models, and utilize Vertex Explainable AI for model evaluation and feature attributions.
- B. Manually download video files from Cloud Storage to local workstations for manual human tagging.
- C. Load all raw binary media directly into Cloud SQL tables and write custom C++ scripts for video transcoding.
- D. Run unmonitored cron jobs on Compute Engine instances that bypass logging and audit trails.

Correct Answer: A

Explanation: Cloud Run functions provide event-driven serverless execution for media processing tasks. Combining foundation models for NLP summarization with Vertex Explainable AI meets the technical requirement that AI systems remain auditable and their decisions explainable.

Topic 2: Core Infrastructure & Reliability Architecture

Question 2.1

An enterprise is designing a mission-critical web application on Google Cloud. According to the Google Cloud Well-Architected Framework's Operational Excellence pillar, how should the operations team establish and track service performance and reliability commitments?

- A. Rely exclusively on vendor marketing claims to guarantee 100% uptime without tracking metrics.
- B. Define SMART Service Level Objectives (SLOs), measure them using Service Level Indicators (SLIs) in Cloud Monitoring, and establish contractual Service Level Agreements (SLAs).
- C. Configure CPU utilization as the sole metric for application health and send email alerts to all developers whenever CPU exceeds 50%.
- D. Avoid setting performance targets to prevent incurring SLA non-compliance penalties.

Correct Answer: B

Explanation: The Well-Architected Framework recommends defining SMART (Specific, Measurable, Achievable, Relevant, Time-bound) SLOs and tracking them using SLIs via Cloud Monitoring to ensure operational readiness and measure compliance against SLAs.

Question 2.2

An e-commerce system experiences sudden, extreme traffic spikes during promotional flash sales. To prevent cascading system failures and maintain application availability, the architect must design the application for graceful degradation. Which approach aligns with Google Cloud reliability principles?

- A. Increase database instance sizes vertically immediately after an outage occurs.
- B. Implement request throttling, drop excess requests early at the edge, use stateless microservices, and design application components to handle partial errors gracefully.
- C. Queue all incoming requests indefinitely in application memory until worker threads become available.
- D. Disable all autoscaling policies to keep server infrastructure static under heavy load.

Correct Answer: B

Explanation: Designing for graceful degradation requires dropping excess traffic early, enforcing throttling, utilizing stateless architectures for horizontal scalability, and handling partial errors with retries to protect backend components.

Question 2.3

A financial services firm requires guaranteed compute capacity for specialized high-memory VM instances during quarterly risk modeling, while seeking to reduce compute costs for fault-tolerant batch analytics jobs that run overnight. How should the architect structure the capacity plan?

- A. Reserve compute capacity using Future Reservations for the high-memory instances, and use Flex-start VMs (Dynamic Workload Scheduling) for the fault-tolerant batch analytics jobs.
- B. Provision standard Compute Engine instances continuously 24/7 across all environments.
- C. Rely exclusively on Spot VMs for mission-critical quarterly risk modeling.
- D. Deploy all batch workloads on legacy on-premises hardware without cloud burst capacity.

Correct Answer: A

Explanation: Future reservations provide an assurance of compute capacity for critical machine types up to a year in advance. Flex-start VMs leverage Dynamic Workload Scheduling (DWS) to queue batch jobs and significantly reduce compute costs for fault-tolerant workloads.

Question 2.4

A cloud architect is selecting a deployment archetype for an enterprise database that requires global scale, active-active multi-region availability, and strong transactional consistency across multiple continents. Which architecture should be chosen?

- A. A single-zone Compute Engine VM running MySQL with local persistent disks.
- B. A regional Cloud SQL instance with a read replica in a secondary region.
- C. A global deployment archetype utilizing Cloud Spanner for globally distributed relational data with strong consistency.
- D. A Filestore instance shared over NFS across multiple regions.

Correct Answer: C

Explanation: Cloud Spanner provides a fully managed, globally distributed relational database service supporting multi-regional deployment archetypes with high availability and strong transactional consistency.

Question 2.5

Following an unexpected service outage caused by a cascading database connection exhaustion, an SRE team wants to prevent similar incidents in the future. According to the Google Cloud Well-Architected Framework's Operational Excellence pillar, what is the best operational practice?

- A. Identify and reprimand the engineer who deployed the database code change.
- B. Conduct a blameless postmortem to identify root causes, document actionable preventive measures, and share the findings broadly.
- C. Suppress all alerts for database connection thresholds to reduce alert fatigue.
- D. Freeze all software deployments permanently across the organization.

Correct Answer: B

Explanation: Conducting blameless postmortems helps teams understand systemic root causes without assigning individual blame, fostering a culture of continuous operational improvement and preventing incident recurrence.

Topic 3: Networking & Hybrid Connectivity

Question 3.1

An enterprise organization has multiple independent engineering teams, each managing their own Google Cloud projects. The central network administration team must enforce centralized control over subnets, IP address blocks, firewall rules, and VPN gateways while allowing service project admins to deploy resources into shared subnets. Which topology should be implemented?

- A. Create isolated standalone VPC networks in each project and connect them using public IP addresses.
- B. Implement Shared VPC, designating a central administration project as the Host Project and linking team projects as Service Projects.
- C. Set up full-mesh VPC Network Peering between 50 distinct project VPCs without centralized administration.
- D. Place all resources across the company into a single default VPC network in a single project.

Correct Answer: B

Explanation: Shared VPC allows an organization to centralize network administration in a host project while sharing subnets with service projects, enabling strict separation of duties between network admins and service developers.

Question 3.2

An enterprise requires a secure, private connection between its on-premises data center and Google Cloud VPC that supports dynamic BGP routing and guarantees a 99.99% availability SLA. How should the network architect configure the Cloud VPN connection?

- A. Deploy a Classic VPN gateway with a single static IPsec tunnel over the public internet.
- B. Configure a Google Cloud HA VPN gateway with two active interface endpoints and four active IPsec tunnels connected to dual on-premises VPN routers.
- C. Create an open SSH tunnel running on a standard Compute Engine instance.
- D. Use VPC Network Peering across the public internet without IPsec encryption.

Correct Answer: B

Explanation: High Availability (HA) VPN provides a 99.99% availability SLA by utilizing a dual-interface gateway topology with active-active BGP routing and redundant IPsec tunnels across separate failure domains.

Question 3.3

A global corporation operates multiple VPC networks across Google Cloud regions and wants to simplify management of its hub-and-spoke network architecture while seamlessly routing traffic between cloud VPCs and on-premises sites. Which service fulfills this requirement?

- A. Network Connectivity Center (NCC) using hub and spoke resources.
- B. External Application Load Balancers.
- C. Legacy SFTP file gateways.
- D. Static route tables configured manually inside each Compute Engine VM.

Correct Answer: A

Explanation: Network Connectivity Center (NCC) serves as a centralized management hub for connecting hub-and-spoke topologies across Google Cloud VPCs, hybrid WANs, and on-premises networks.

Question 3.4

Compute Engine virtual machines deployed in a private subnet without external IP addresses need to access Google Cloud APIs (such as Cloud Storage and BigQuery) securely without traversing the public internet. What feature should be enabled?

- A. Assign public IPv4 addresses to all virtual machines.
- B. Enable Private Google Access on the subnet.
- C. Create an external HTTP Load Balancer with an open ingress rule.
- D. Deploy an unencrypted NAT gateway instance running open proxy software.

Correct Answer: B

Explanation: Private Google Access allows VM instances with internal IP addresses only to reach Google Cloud APIs and services privately over internal Google network routing.

Question 3.5

A public-facing web application deployed behind a global Application Load Balancer is targeted by distributed denial-of-service (DDoS) attacks and SQL injection attempts. Which edge security control should be implemented?

- A. Configure Google Cloud Armor security policies at the edge to inspect traffic and enforce rate limiting and WAF rules.
- B. Apply SSH key restrictions using OS Login on worker instances.
- C. Enable VPC Flow Logs to block incoming malicious packets automatically.
- D. Restrict Cloud KMS key rotation schedules.

Correct Answer: A

Explanation: Google Cloud Armor provides edge DDoS protection and Web Application Firewall (WAF) rule enforcement for global Application Load Balancers to defend against web attacks and volumetric traffic spikes.

Topic 4: Cloud Security, Identity & Governance

Question 4.1

An enterprise uses an external Identity Provider (IdP) for employee authentication. Applications running in an on-premises datacenter need to authenticate to Google Cloud APIs to access storage buckets. The security team prohibits downloading and storing long-lived service account JSON keys. What mechanism should be implemented?

- A. Workload Identity Federation using short-lived tokens federated with the external IdP.
- B. Generate service account private keys, commit them to Git repositories, and sync them to on-premises servers.
- C. Make the Cloud Storage buckets publicly readable over the internet.
- D. Assign the roles/owner IAM primitive role to all application users.

Correct Answer: A

Explanation: Workload Identity Federation allows external workloads (on-premises or multicloud) to exchange credentials from an external IdP for short-lived Google Cloud IAM tokens, eliminating the risks of long-lived service account keys.

Question 4.2

Administrators need administrative SSH access to Compute Engine instances located in private subnets without external IP addresses. The security policy requires central IAM access control and multi-factor authentication (MFA) without exposing SSH ports to the internet. Which combination of features meets these security requirements?

- A. Enable OS Login on instances and use Identity-Aware Proxy (IAP) TCP forwarding.
- B. Attach public IP addresses to all VMs and distribute static SSH keys via email.
- C. Open port 22 on default firewalls to IP range 0.0.0.0/0.
- D. Embed root password credentials in custom boot scripts.

Correct Answer: A

Explanation: OS Login ties VM SSH access to centralized IAM permissions and supports MFA. IAP TCP forwarding allows authorized users to establish secure SSH sessions to VMs without external IP addresses over encrypted tunnels.

Question 4.3

A healthcare firm wants to ensure that internal employees cannot copy or transfer sensitive patient data from BigQuery datasets to unauthorized personal Cloud Storage buckets outside the organization, even if the employees possess valid IAM admin permissions. Which security boundary must be established?

- A. Create a VPC Service Controls perimeter encompassing BigQuery and Cloud Storage services.
- B. Deploy an internal load balancer with round-robin health checks.
- C. Enable Cloud Trace on all BigQuery API calls.
- D. Increase the billing quota on BigQuery queries.

Correct Answer: A

Explanation: VPC Service Controls create security perimeters around Google Cloud resources and service APIs to prevent data exfiltration, ensuring data cannot be copied to unauthorized external resources even by compromised credentials.

Question 4.4

A financial services firm handling sensitive cardholder data requires encryption keys for data at rest to be managed in a hardware security module (HSM) with automated key provisioning and regular key rotation to comply with regulatory standards. Which key management strategy should be adopted?

- A. Rely solely on default Google-managed encryption keys without key rotation visibility.
- B. Generate Customer-Managed Encryption Keys (CMEK) using Cloud Key Management Service (Cloud KMS) with HSM protection and KMS Autokey.
- C. Store raw, unencrypted encryption keys in hardcoded configuration files inside application source code.
- D. Disable encryption at rest across all storage buckets.

Correct Answer: B

Explanation: Cloud KMS with Customer-Managed Encryption Keys (CMEK) and HSM protection provides hardware-backed key security, automated rotation, and compliance auditing for sensitive enterprise workloads.

Question 4.5

A security team needs to enforce organizational guardrails that prohibit any developer from creating VM instances with external IP addresses across all projects under a specific folder. How should this restriction be configured?

- A. Write custom Python scripts that run hourly to delete VMs with external IPs.
- B. Define an Organization Policy constraint (compute.vmExternalIpAccess) at the folder level.
- C. Delete all default VPC networks manually once per week.
- D. Remove all billing accounts from developer projects.

Correct Answer: B

Explanation: The Organization Policy Service allows centralized administrators to set declarative security constraints across resource hierarchies (Organization, Folders, Projects), such as disabling external IP addresses.

Topic 5: Data Engineering, Analytics & BigQuery ML

Question 5.1

A retail company needs to replicate real-time transactional updates from an operational PostgreSQL database into a BigQuery data warehouse with minimal latency for analytics. Which serverless service handles this Change Data Capture (CDC) replication pattern?

- A. Datastream.
- B. Cloud Storage FUSE.
- C. Compute Engine custom cron scripts.
- D. Cloud DNS.

Correct Answer: A

Explanation: Datastream is a serverless Change Data Capture (CDC) and replication service that streams real-time database updates from source databases (like PostgreSQL) directly into BigQuery.

Question 5.2

A data engineering team manages hundreds of complex SQL queries in BigQuery. They need to build an ELT pipeline that supports version control, dependency management, SQL testing, and automated orchestration. Which tool is designed for this SQL workflow?

- A. Dataform.
- B. Filestore.
- C. Memorystore for Redis.
- D. Cloud Interconnect.

Correct Answer: A

Explanation: Dataform enables data teams to develop, version-control, test, and execute SQL workflows directly within BigQuery, turning raw data into organized ELT data pipelines.

Question 5.3

An IoT platform receives streaming telemetry from thousands of connected vehicles. The data architect must implement an end-to-end pipeline to stream high-volume events, transform payloads in real time, and load results into BigQuery to power real-time dashboards. Which stack should be chosen?

- A. Cloud Pub/Sub, Cloud Dataflow (Apache Beam), and BigQuery.
- B. Manual SFTP file upload to Compute Engine local disks.
- C. Exporting CSV logs via email to analytical teams.
- D. Direct batch processing using SQL Server Management Studio once a month.

Correct Answer: A

Explanation: Cloud Pub/Sub ingests high-volume streaming telemetry, Cloud Dataflow executes scalable stream processing using Apache Beam, and BigQuery serves as the real-time analytical sink for dashboards.

Question 5.4

A data analyst team wants to train and evaluate predictive machine learning models directly on multi-terabyte datasets stored in BigQuery without exporting data or writing Python/TensorFlow code. What feature should they use?

- A. BigQuery ML (BQML) using standard SQL statements.
- B. Exporting data to local desktop spreadsheets for analysis.
- C. Deploying manual C++ algorithms on custom GKE worker nodes.
- D. Cloud Run functions running basic shell scripts.

Correct Answer: A

Explanation: BigQuery ML (BQML) enables users to create, evaluate, and run inference on predictive machine learning models directly inside BigQuery using standard SQL queries.

Question 5.5

An enterprise organization wants to securely publish, discover, and share curated BigQuery datasets with external business partners without copying or moving the underlying physical data. Which BigQuery feature provides this capability?

- A. Analytics Hub.
- B. Cloud VPN HA gateways.
- C. Filestore NFS network shares.
- D. Exporting raw database dumps to public HTTP servers.

Correct Answer: A

Explanation: Analytics Hub allows organizations to publish and subscribe to shared data products securely within BigQuery without copying underlying data assets.

Topic 6: AI & Machine Learning Infrastructure

Question 6.1

An AI research team is training large-scale generative AI foundation models requiring thousands of TPU and GPU chips, high-speed interconnects, and scalable parallel storage. Which integrated infrastructure platform is optimized for these heavy AI training workloads?

- A. AI Hypercomputer architecture combining TPUs, GPUs, high-performance storage, and cluster networking.
- B. Single-zone N1 Compute Engine VMs with standard HDD disks.
- C. Filestore Basic tiers over public internet links.
- D. Shared VPC host projects running legacy Web logic.

Correct Answer: A

Explanation: The AI Hypercomputer is Google Cloud's supercomputing architecture integrating hardware (TPUs, GPUs), high-performance storage (Cloud Storage FUSE / Managed Lustre), and optimized cluster networking for AI workloads.

Question 6.2

A company wants to train a custom predictive machine learning model using tabular dataset files stored in Cloud Storage. They lack dedicated ML engineers and require automated feature engineering, hyperparameter tuning, and model deployment. What Vertex AI capability should they choose?

- A. Vertex AI (Agent Platform) AutoML.
- B. Custom PyTorch training on unmanaged Compute Engine instances.
- C. Writing custom C++ matrix multiplication routines from scratch.
- D. Executing basic BASH scripts in Cloud Shell.

Correct Answer: A

Explanation: Vertex AI AutoML automatically performs data preprocessing, feature engineering, model architecture selection, hyperparameter tuning, and model deployment for tabular data.

Question 6.3

A bank deploys a machine learning model to evaluate loan application risks. To comply with financial regulations, the bank must explain which input features (e.g., credit score, income) contributed most significantly to each individual credit prediction. Which feature provides these attributions?

- A. Vertex Explainable AI.
- B. Cloud Logging export filters.
- C. VPC Flow Logs.
- D. Google Cloud Armor edge rules.

Correct Answer: A

Explanation: Vertex Explainable AI integrates with BigQuery ML and Vertex AI models to provide feature-based explanations and attributions, clarifying how input parameters influence model decisions.

Question 6.4

An enterprise security officer is establishing security guardrails for AI development. They want to adopt Google's framework for securing AI ecosystems, supply chains, and data pipelines against tampering and prompt threats. Which framework should be implemented?

- A. Secure AI Framework (SAIF).
- B. PCI-DSS compliance for GKE.
- C. Basic Linux file permission bits.
- D. Sustained Use Discounts.

Correct Answer: A

Explanation: Google's Secure AI Framework (SAIF) provides a structured methodology and set of core security principles designed to secure AI pipelines, model artifacts, supply chains, and deployment systems.

Question 6.5

A mobile application developer needs to incorporate sentiment analysis and entity recognition into an app without training custom machine learning models. Which Google Cloud development path should they select?

- A. Pre-trained ML APIs (such as the Cloud Natural Language API).
- B. Custom C3 AI deployment on bare metal servers.
- C. Building custom deep learning neural networks in C++.
- D. Manual human evaluation using email forms.

Correct Answer: A

Explanation: Pre-trained ML APIs (like the Cloud Natural Language API) provide out-of-the-box REST API capabilities for entity extraction and sentiment analysis without requiring custom model training.

Topic 7: Serverless, Containers & DevOps Automation

Question 7.1

A software company is modernizing a stateless web application. They want to deploy containerized code that automatically scales instance counts based on web traffic demand—including scaling down to zero when idle—without managing underlying VM infrastructure. Which service should be selected?

- A. Cloud Run.
- B. Compute Engine standalone virtual machines.
- C. Filestore High Performance tier.
- D. Cloud DNS public zones.

Correct Answer: A

Explanation: Cloud Run is a fully managed serverless platform that runs stateless containerized applications, offering automated scaling based on incoming traffic requests including scale-to-zero.

Question 7.2

A DevOps engineering team needs to provision, manage, and version-control Google Cloud infrastructure resources declaratively across multiple environments using reusable infrastructure modules. Which tool meets this requirement?

- A. Terraform.
- B. Manual shell commands in Cloud Shell.
- C. Imperative Python scripts modifying console DOM elements.
- D. Microsoft SQL Server Management Studio.

Correct Answer: A

Explanation: Terraform is an Infrastructure as Code (IaC) tool that allows teams to declaratively provision, manage, and modularize Google Cloud resources using version-controlled configurations.

Question 7.3

An enterprise wants to build a continuous integration and continuous deployment (CI/CD) pipeline for microservices. The solution must automate container image builds, execute automated tests, and manage progressive application releases across staging and production with rollback support. Which combination of services fulfills this pipeline?

- A. Cloud Build for automated container builds and Cloud Deploy for continuous delivery releases and rollbacks.
- B. Cloud Storage FUSE with manual FTP uploads.
- C. Local developer desktop scripts triggering email notifications.
- D. Direct production database updates using ad-hoc SQL queries.

Correct Answer: A

Explanation: Cloud Build provides fully managed continuous integration (CI) for building and testing container images, while Cloud Deploy automates continuous delivery (CD) across target environments with built-in release promotion and rollback capabilities.

Question 7.4

A DevSecOps team must enforce container software supply chain security for GKE clusters. They require container images to be automatically scanned for vulnerabilities during build time, and GKE must block the deployment of any container that lacks an official approval attestation. Which services fulfill these requirements?

- A. Artifact Analysis for vulnerability scanning and Binary Authorization for enforcing deployment attestations.
- B. Cloud Logging export sinks and Cloud Trace.
- C. Spot VMs and Sustained Use Discounts.
- D. Basic HTTP firewalls with open egress.

Correct Answer: A

Explanation: Artifact Analysis automatically scans container images for vulnerabilities, and Binary Authorization enforces deploy-time policy checks to ensure only signed, verified container attestations can run on GKE.

Question 7.5

A developer needs to build an asynchronous system where uploading an image file to a Cloud Storage bucket automatically triggers a background task to process the file and send an event notification. Which serverless architecture is best suited for this task?

- A. Use Cloud Storage event notifications via Eventarc to trigger a serverless Cloud Run function.
- B. Run a continuous polling loop inside a dedicated 24/7 Compute Engine VM.
- C. Require users to submit a manual support ticket via an IVR call center.
- D. Store files inside local browser cache memory.

Correct Answer: A

Explanation: Combining Cloud Storage event triggers, Eventarc routing, and serverless execution via Cloud Run functions creates an asynchronous, event-driven pipeline that runs code automatically in response to file events.

Topic 8: Observability & FinOps

Question 8.1

A cloud operations team needs a unified platform to collect metrics, aggregate application logs, track distributed request latency across microservices, and detect runtime code errors. Which suite of tools should be used?

- A. Google Cloud Observability (Cloud Monitoring, Cloud Logging, Cloud Trace, and Error Reporting).
- B. Local plain-text log files saved on individual developer laptops.
- C. Unmonitored CSV log exports sent via email.
- D. Cloud KMS Autokey.

Correct Answer: A

Explanation: Google Cloud Observability provides a unified platform incorporating Cloud Monitoring for metrics, Cloud Logging for audit trails/logs, Cloud Trace for request latency analysis, and Error Reporting for application exception tracking.

Question 8.2

A FinOps practitioner wants to analyze granular cloud resource spending patterns across projects and departments. They need to query raw usage data using SQL and build custom cost visualization dashboards. What setup should be implemented?

- A. Export Cloud Billing data continuously to a BigQuery dataset and visualize spending using labels and custom dashboards.
- B. Manually download paper credit card statements at the end of every quarter.
- C. Rely on email notifications when compute instances fail.
- D. Estimate monthly spend using custom spreadsheets.

Correct Answer: A

Explanation: Exporting Cloud Billing data to BigQuery enables FinOps teams to run detailed SQL queries, filter spend using resource labels/tags, and build custom dashboards for cost allocation and trend analysis.

Question 8.3

An enterprise runs predictable baseline database workloads on Compute Engine virtual machines 24 hours a day, 365 days a year. How can the FinOps team optimize compute spending for these predictable workloads without changing instance machine types?

- A. Purchase Committed Use Discounts (CUDs) for a 1-year or 3-year term.
- B. Deploy all production database workloads on Spot VMs.
- C. Shut down production databases every night.
- D. Pay default on-demand rates without discount commitments.

Correct Answer: A

Explanation: Committed Use Discounts (CUDs) offer significant pricing discounts in exchange for a commitment to maintain continuous resource usage over a 1-year or 3-year period, making them ideal for predictable baseline workloads.

Question 8.4

A GKE cluster administrator notices that cluster nodes are frequently underutilized, while some application Pods experience resource starvation due to inaccurate CPU and memory allocation settings. How should cluster autoscaling be configured to optimize costs and performance?

- A. Combine Cluster Autoscaler, Horizontal Pod Autoscaler (HPA), and Vertical Pod Autoscaler (VPA).
- B. Disable all autoscaling mechanisms and hardcode fixed node counts.
- C. Deploy all application Pods in a single master node pool without resource limits.
- D. Delete cluster worker nodes manually whenever memory usage drops.

Correct Answer: A

Explanation: Combining Cluster Autoscaler (adjusting node counts), Horizontal Pod Autoscaler (scaling Pod replicas), and Vertical Pod Autoscaler (right-sizing Pod CPU/memory requests) optimizes GKE resource utilization and cost efficiency.

Question 8.5

An organization wants to prevent unexpected cloud bill surges. The finance department needs automated alerts to notify team leads when monthly project spending reaches 50%, 80%, and 100% of their budgeted limit. Which cost management feature fulfills this need?

- A. Cloud Billing budgets and Pub/Sub automated alert notifications.
- B. Disabling Cloud Logging across all projects.
- C. Setting Organization Policy constraints to delete projects after 30 days.
- D. Requesting quota decreases on all Compute Engine APIs.

Correct Answer: A

Explanation: Cloud Billing budgets allow organizations to define financial targets and configure automated alerts (via email or Pub/Sub) when actual or forecasted spend reaches specified percentage thresholds.